Nodes	Class	Hamiltonian Form	Compact	Extended	Harmonic	#Circuits
2	1	$n_1^2 + \cos(\theta_1)$	1	0	0	2
	2	$q_1^2 + \varphi_1^2 + \cos(\varphi_1)$	0	1	0	2
3	1	$n_1^2 + q_2^2 + \varphi_2^2 + \cos(\theta_1)$	1	0	1	2
	2	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1)$	1	0	1	4
	3	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	2
	4	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	5	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	6	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	7	$n_1^2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2)$	2	0	0	3
	8	$n_1^2 + n_1 n_2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2)$	2	0	0	3
	9	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	3
	10	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	3
	11	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	7
	12	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	7
	13	$n_1^2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2)$	1	1	0	4
	14	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2)$	1	1	0	4
	15	$n_1^2 + n_1 n_2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2) + \cos(\theta_1 + \theta_2)$	2	0	0	4
	16	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\theta_1 + \varphi_2)$	1	1	0	6
	17	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2) + \cos(\varphi_1 + \varphi_2)$	0	2	0	6
	18	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2) + \cos(\varphi_1 + \varphi_2)$	0	2	0	4
	19	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2) + \cos(\theta_1 + \varphi_2)$	1	1	0	6